

CivicMate: An AI-Driven City Grievances

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Abstract: Rapid urbanization in India has intensified many passive civic issues like potholes, garbage dumping, broken streetlights, and water leakage in public spaces. These problems are noticeable, but often remain unsolved for a long time due to slow and complicated traditional grievance redressal system. Citizens frequently face long forms, unclear complaint status, and uncertainty about whether their reports reach the correct authorities. To address these challenges, we propose CivicMate, a web-based smart grievance platform that simplifies issue reporting and enhances coordination between citizens and local bodies. Users can upload a photo, provide a short description, and the system automatically detects their location. The platform employs artificial intelligence to verify image appropriateness and recommend the correct issue category, enabling quicker routing to relevant departments. After that, the complaints are automatically ranked and forwarded to the appropriate authorities. Citizens can track complaint progress in real time, receive SMS updates, and engage with community-reported issues through upvotes. On the other side, the administrative dashboard supports municipal staff in efficient complaint management, task assignment, and pattern analysis across reported issues. Ultimately, CivicMate facilitates clear communication between citizens and authorities, leading to quicker solutions and greater transparency. This entire system aims to build cleaner, safer, and smarter Indian cities while strengthening public trust in local governance.

Keywords— Smart Cities, Grievance Redressal, CivicTech, Artificial Intelligence, Deep Learning, CNN, Image Classification, NSFW Detection, E-Governance, Urban Infrastructure, Complaint Tracking, Transparent Governance, Data Confidentiality.

I. INTRODUCTION

India's rapid urban expansion, now encompassing over 400 million citizens continue to put heavy pressure on essential civic infrastructure. Residents routinely encounter everyday issues such as uncollected garbage, damaged streetlights, dangerous potholes, and persistent water leakage, all of which significantly impact their quality of living conditions. Although laudable government initiatives such as the Swachh Bharat Mission and Namami Gange aim to enhance urban environments, many complaints regrettably remain unresolved for extended periods. This is largely due to the limitations of traditional grievance systems that depend on phone calls, emails, or physical visits, making them slow, inconvenient, and lacking any dependable mechanism for complaint tracking. As a result, residents are often uncertain about whether their issues are being acted upon, while municipal authorities struggle to process large volumes of unorganized reports and find it difficult to prioritize or analyze urgent urban problems. In many cases, complaints are misplaced, delayed, or addressed without proper verification, further weakening citizen trust.

To directly tackle these long-standing challenges, we introduce CivicMate, a web-based intelligent platform designed to simplify and modernize the grievance redressal process. The system ensures reliability and accountability through secure user authentication, after which citizens can submit complaints by providing a short description and a crucial accompanying photo. The system further reduces user effort by automatically detecting the complainant's location, removing the need for manually entering address details and enabling faster, more accurate issue reporting.

The moment when user submits their complaint, platform's AI engine immediately springs into action. It swiftly processes the information, first checking the uploaded image to filter inappropriate content using a detection model. Following this, it automatically assigns the complaint to the relevant category for quicker routing to the appropriate department. Throughout the process, users remain fully informed and can monitor the status of their complaint in real time, similar to tracking an online order, and receive SMS updates whenever the status changes. Furthermore, the platform allows citizens to view issues reported by others in their locality and support them through upvotes, helping highlight community priorities for authorities.

On the administrative side, CivicMate provides a comprehensive dashboard specifically for municipal authorities and their staff. This centralized interface helps officials easily review incoming complaints, spot recurring problem areas, and assign tasks to the appropriate team members. Officials can also update each complaint's status—from "In Progress" to "Resolved" ensuring clear visibility throughout the workflow. These features make it easier for prioritizing urgent issues and prevent any complaint from being overlooked.

Overall, CivicMate brings much-needed transparency, speed, and reliability to the grievance redressal process. By automating complaint verification, category prediction, and easy status tracking, the platform allows citizens to report issues without unnecessary hassle and receive faster, more accountable responses. It serves as an important step toward building cleaner, safer, and smarter, while strengthening trust between communities and local government authorities.

II. PROBLEM STATEMENT

In India, as cities are growing rapidly, citizens encounter a wide range of civic issues on a daily basis such as overflowing garbage bins, broken streetlights, potholes, open manholes, and polluted public spaces. These problems not only affect the appearance of the city but also pose significant health and safety risks for residents. Although several

government initiatives aim to address such challenges, the methods available for citizens to report these issues remain largely outdated, slow, and ineffective, limiting timely intervention and proper resolution.

Most existing grievance redressal systems are difficult to use and burden citizens with complicated procedures. This often involves navigating lengthy forms, making time-consuming visits to municipal offices, or engaging with centralized portals like CPGRAMS, which frequently suffer from a critical lack of real-time tracking and transparency, offering no reliable assurance that a complaint will even reach the correct department. As a result, many complaints simply vanish into delays, overlooked, or incorrectly marked as resolved without proper action.

A further significant challenge lies in the unstructured manner in which complaints are often submitted and processed. Many citizens express their concerns on informal channels such as social media, resulting in large amounts of scattered and unverified information for authorities to sort through. This makes it incredibly difficult to confirm the authenticity or practical utility of these reports. Additionally, most local government departments lack advanced analytical tools to track complaint trends, monitor resolution timelines, or identify areas that require immediate attention. Due to this absence of data-driven insights, authorities struggle to plan targeted interventions and address problems efficiently.

CivicMate addresses these problems by offering a simple and user-friendly platform where citizens can report issues by describing the problem, uploading a photo, and allowing the system to automatically detect their location. The platform uses AI to verify the image content and recommend the appropriate complaint category. Authorities receive a clear and organized dashboard that helps them review, prioritize, and manage complaints based on urgency and type. Meanwhile, citizens can track the progress of their submissions in real time and receive SMS updates once their issues are resolved. This integrated approach makes the grievance redressal process smarter, faster, and more transparent, ultimately contributing to cleaner, safer, and more efficient urban environments.

III. CONTRIBUTION

Our objective is to develop a digital platform that minimizes irregularities in reporting public issues and encourages citizens to use civic resources effectively. CivicMate is not just another complaint-filing system; it introduces meaningful improvements in how residents and city authorities interact.

- A. Citizens can monitor the progress of their complaints, reducing uncertainty and increasing confidence that their issues are being addressed.
- B. Strengthening data security using AI-based anonymization techniques help to create a secure environment where people can report sensitive issues without compromising their privacy.
- C. The system uses image processing and CNN models to detect and filter inappropriate or NSFW content before it reaches authorities.
- D. An AI-powered chatbot assists users throughout the reporting process, making the experience more intuitive and encouraging greater participation.

IV. LITERATURE REVIEW

Several research efforts have examined digital platforms designed to support civic engagement and improve complaint management. This section reviews the most relevant systems and approaches, highlighting their strengths as well as their limitations.

S. Raj and A. Kumar proposed a digital grievance system that applies AI and Natural Language Processing (NLP) to automatically categorize text-based complaints [1]. Their approach reduces manual effort by directing issues to the appropriate departments and offering an administrative dashboard for monitoring. However, the system lacks support for image-based submissions, which limits the contextual understanding of reported issues. It also does not provide features for citizen interaction, such as viewing or tracking complaints submitted by others, which restricts community engagement and overall transparency.

A. Dubey et al. demonstrated the potential of deep learning to analyze how people perceive urban environments by processing street-level images [2]. Their model could predict subjective qualities like safety, showing that AI can effectively interpret visual data from a city environment. This insight supports our use of image classification to recommend complaint categories in CivicMate. However, their work was focused on perception analysis and did not extend to building an interactive or operational grievance management system.

J. Zhang et al. developed the FAKEDETECTOR system, which uses a neural network to identify fake news by analysing text content and user profiles [3]. The idea of verifying information through AI aligns with our approach to filtering misleading or false complaints. However, their solution is heavily text-focused and computationally demanding. This highlights the need for a lighter, more practical method that can also work with images, which is essential for validating the visual evidence included in civic issue reports.

S. Kumar et al. proposed an innovative approach using Scene Graph Models to interpret civic issues directly from images [4]. Their system can detect relationships between objects—for example, identifying a “pothole near a streetlight” which offers far more context than a basic category label. Although this level of detail is valuable, the method depends on a small dataset and requires substantial computational resources, making it difficult to use in real-time applications or deploy at a city-wide scale.

The “Snap Send Solve” application, widely used in Australia and New Zealand, provides a simple way for citizens to report local issues directly to municipal councils [5]. Its major strength lies in its user-friendly interface, which makes capturing and submitting complaints quick and easy. However, the platform does not offer users any direct feedback or a way to track the status of their reports—an essential feature that CivicMate specifically addresses to enhance transparency and user engagement.

An AI-based public complaint chatbot has also been explored to guide users step-by-step through the reporting process [6]. This work demonstrates the practicality of using conversational interfaces to collect complaint details in a simple manner. The authors recommend enhancements such as multilingual support and communication features between administrators and supervisors. These ideas are valuable for

improving accessibility and operational efficiency, and they align well with the goals of our system.

Other web-based grievance platforms, such as the system described by S. Suvetha et al. [7], require citizens to manually choose the problem category and enter their location. This manual process is time-consuming, inconvenient, and can lead to incorrect or inconsistent submissions. Furthermore, the system has difficulty filtering out complaints that fall outside its operational area and offers no reliable way to verify the authenticity of text-only reports. These limitations make it challenging for authorities to manage and respond to complaints efficiently.

Y. Shrinewar et al. proposed a system that allows users to submit an image along with a manually selected category [8]. Although this represents an improvement over text-only systems, the solution is limited to a single city and still depends heavily on the user to correctly classify the issue. This underscores the need for an automated and scalable platform that can intelligently process complaints across different regions while reducing the manual effort required from citizens.

Komar and Ersoy explored how IEEE 802.11 WLAN networks can be used to estimate a user's location and support location-based services [9]. Their study demonstrates that Wi-Fi infrastructure can provide reasonably accurate positioning without the need for additional hardware, making it a practical solution for indoor and urban environments. However, their work is limited to the technical aspects of localization and does not address the broader challenges of civic issue reporting, such as complaint management, verification, or workflow automation.

Chi et al. introduced a notification framework aimed at improving tracking and timely communication within digital systems [10]. Their work emphasizes how real-time alerts can significantly enhance user engagement by keeping individuals informed about the status of ongoing activities or requests. The study demonstrates the importance of delivering quick and reliable updates to maintain transparency and user satisfaction. This system supports use of SMS and dashboard notifications, which keep citizens aware of their complaint progress at every stage. However, their research focuses on generic tracking mechanisms and does not address the end-to-end workflow or complexity of managing civic grievances.

Do and Hussain introduced a hybrid framework for personalizing cloud-based e-governance services, aiming to deliver more citizen-centric digital experiences [11]. Their approach combines different computational techniques to tailor services based on user behavior and preferences, ultimately improving accessibility and satisfaction. However, their work focuses on general e-governance personalization rather than managing civic complaints or integrating AI-driven verification features.

Soomro et al. presented an in-depth review of big data analytics in smart city applications, highlighting how large-scale urban data can support better decision-making and resource management [12]. Their work emphasizes the value of data-driven insights for improving city operations, which aligns with CivicMate's use of complaint trends and analytics for municipal planning. However, their study focuses on broader smart city data frameworks rather than AI-assisted grievance reporting and resolution.

V. PROPOSED METHODOLOGY

The methodology used in CivicMate focuses on developing a centralized, intelligent, and scalable grievance redressal system that places citizens at the center of the process. The approach is built around three key components: structured data collection, AI/ML-driven processing, and fast, transparent complaint resolution. Together, these components ensure the accuracy of reported issues, enable automatic categorization and prioritization for efficient handling, and maintain accountability throughout the complaint lifecycle. This integrated methodology strengthens the overall effectiveness of urban governance.

A. System Architecture

The CivicMate platform is built on a robust, multi-layered architecture designed to ensure seamless operation, strong security, and scalability for urban grievance redressal. This structured design, illustrated in Fig. 1, clearly defines the responsibilities of each layer, enabling smooth transitions from citizen complaint submission to administrative processing and resolution.

1) *User Interface (UI) Layer*: This layer serves as the primary interaction point for both citizens and city officials through a responsive web application. Citizens can quickly submit complaints by uploading images, providing short descriptions, and allowing the system to auto-detect their location. They can also track the status of their reports in real time and receive timely updates. For administrators, the UI offers clear tools to review, filter, and manage incoming complaints, enabling them to understand ongoing issues and respond more effectively.

2) *Middleware Layer*: This layer functions as the intelligent core of the system, validating user inputs and enforcing security protocols. It manages the platform's AI-driven capabilities, including image content verification and automatic complaint categorization, ensuring each issue is routed to the appropriate department. Additionally, the middleware handles secure API communication, enabling smooth and reliable data exchange between the user interface and the database. This coordination ensures the overall efficiency and correctness of the system's operations.

3) *Database Layer*: This layer forms the foundation of CivicMate and is implemented using MongoDB for secure and flexible data storage. It maintains all essential information, including citizen and administrator profiles, detailed complaint records such as images, locations, and AI-generated categories, as well as complete status histories. The stored data also supports administrative analytics by providing insights into complaint patterns and resolution performance, enabling more informed and data-driven decision-making for city authorities.

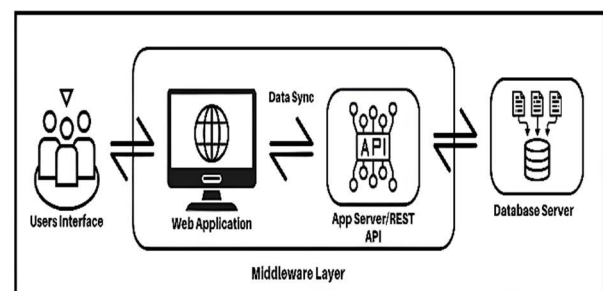


Fig. 1 System Architecture

B. System Workflow

This platform follows a streamlined, AI-driven workflow that improves both citizen participation and administrative efficiency. As illustrated in Fig. 2, Fig. 3, and Fig. 4, the system ensures that every complaint is processed in a clear and organized manner—from the moment it is submitted to its final resolution. The workflow is designed to minimize manual effort, reduce delays, and provide full transparency, making it easier for users to report issues and for authorities to manage them effectively.

1) *Citizen Interaction & Submission*: As shown in Fig. 2, the workflow begins with secure user authentication, after which citizens access a simple dashboard to file complaints. They describe the issue, upload a photo, and rely on automatic location detection to avoid manually entering address details. The system's AI engine quickly moderates the uploaded image to ensure that inappropriate content is filtered out. Once submitted, each complaint is assigned a unique tracking ID, allowing users to monitor its progress in real time and receive instant notifications as the status changes. Citizens can also view reports submitted by others in their area and support them through upvotes, helping authorities identify and prioritize issues that affect the larger community. This intuitive process encourages active participation while ensuring accuracy and transparency throughout the reporting journey.

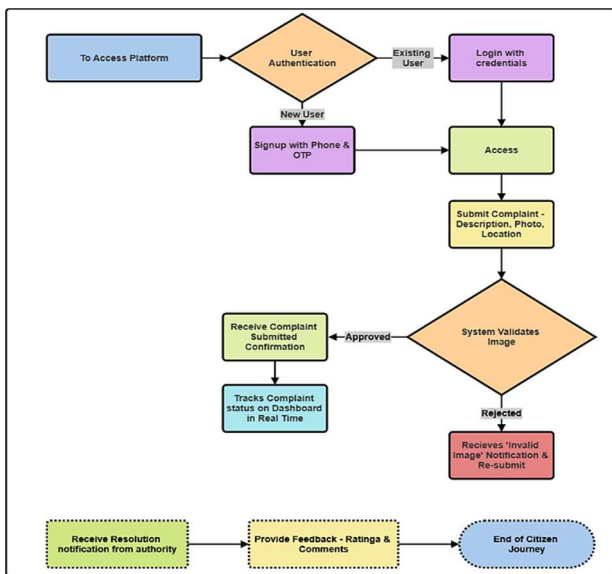


Fig. 2 Citizen-Centric Workflow

2) *Authority Management & Resolution*: As shown in Fig. 3, the Admin Module provides municipal authorities with a centralized dashboard where they can view, organize, and respond to all incoming complaints. Staff members can update the status of each case, assign tasks to field teams, add remarks, and review citizen feedback after resolution. The platform also includes analytics tools that help officials identify recurring issues, monitor workload distribution, and optimize resource allocation. In addition, the system highlights high-priority complaints—such as safety hazards or time-sensitive issues—allowing administrators to address critical cases first. It also supports departmental collaboration by enabling multiple officials to track, comment on, or escalate complex complaints when required. This structured and transparent approach helps authorities ensure timely responses and prevents any complaint from being missed or mishandled.

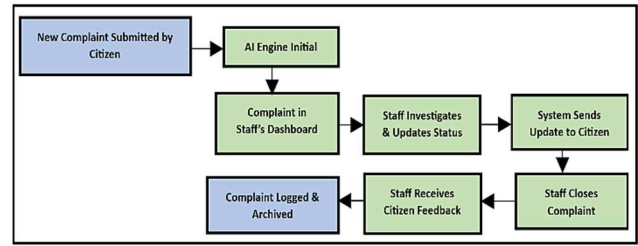


Fig. 3 Admin-Centric Workflow

3) *Machine Learning Model for NSFW Detection*: Automated visual content moderation has become essential for modern digital platforms, and frameworks like NudeNet play a key role in ensuring safety and compliance. As shown in Fig. 4, an open-source solution flow, NudeNet offers transparency and local deployability, making it suitable for academic research and privacy-focused environments where proprietary cloud APIs fall short. The framework has progressed from basic CNN classifiers to advanced object-detection models such as RetinaNet and YOLOv8, resulting in significantly improved precision and recall as well as faster real-time inference through ONNX runtimes. Despite these strengths, its performance still depends heavily on the quality and ethical sourcing of training data, and it remains less effective than newer multimodal AI systems when evaluating subtle or suggestive visual content.

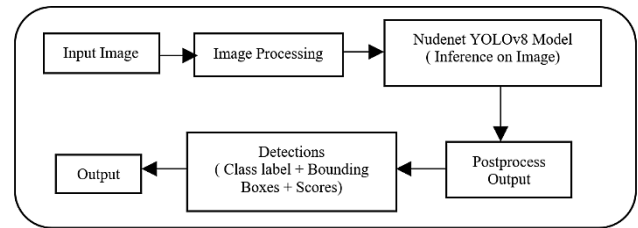


Fig. 4 NSFW Detection Model

4) *High-Level System Flow*: As Fig. 5 illustrates the overall flow of information within CivicMate, connecting both citizen-side actions and administrative processes. Once a complaint is submitted, the system performs a series of automated checks including image validation, category prediction, and location tagging before storing the data securely in the backend. The middleware then routes the complaint to the appropriate department based on its type and priority. Administrators receive real-time updates and can assign tasks, modify statuses, and add remarks as the issue progresses. Throughout the process, the system logs every action, ensuring a complete audit trail. Citizens receive notifications at key stages, such as review, assignment, and resolution. This streamlined and interconnected flow ensures transparency, reduces delays, and allows authorities to manage civic grievances more efficiently.

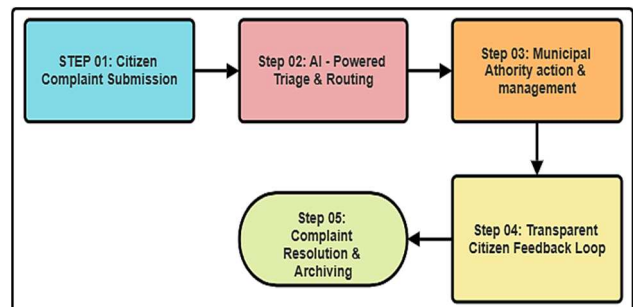


Fig. 5 High-Level System Flow

VI. RESULTS AND DISCUSSION

The results and discussion section presents the practical implementation and performance of the CivicMate platform, supported by key interface screenshots shown in Fig. 6 onward. These figures illustrate the complete user journey from secure authentication and complaint submission to AI-driven processing and real-time status tracking. The visual demonstrations highlight the system's core features, including automated classification, NSFW content detection, chatbot assistance, and community engagement tools. Together, these results validate the platform's ability to streamline grievance reporting and enhance transparency.

A. Secure User Authentication

As Fig. 6 shows the authentication interface, which forms the foundation of CivicMate's secure and user-friendly access system. The platform offers a simple and seamless login and registration process, ensuring strong security without burdening users with complex steps. This streamlined approach lowers entry barriers, allowing citizens to access the platform quickly and begin reporting issues with ease.

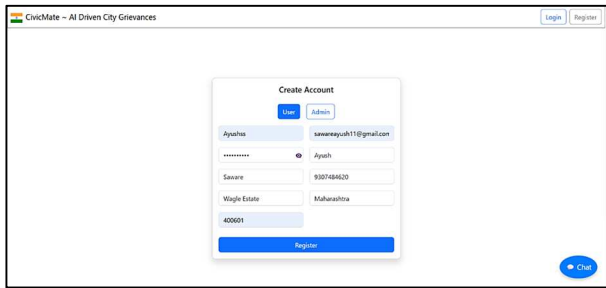


Fig. 6 Authentication

B. Complaint Registration

Once authenticated, citizens can conveniently report civic issues using the complaint registration form shown in Fig. 7. The interface enables users to enter a brief description of the problem, upload a supporting photo, and rely on automatic location detection to enhance accuracy. The layout is simple and intuitive, allowing complaints to be submitted within seconds. By reducing unnecessary steps and simplifying data entry, the platform encourages higher participation and ensures that essential information is captured consistently. This streamlined process forms a crucial part of CivicMate's user-friendly design philosophy.

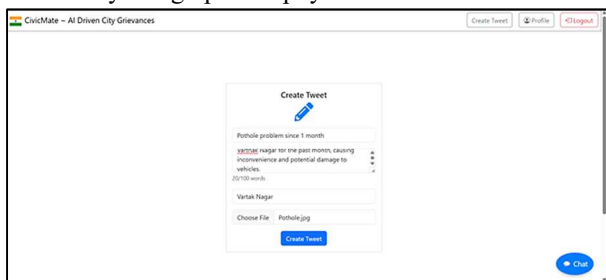


Fig. 7 Complaint Registration

C. NSFW Content Detection

A key security and moderation component of CivicMate is its NSFW (Not Safe for Work) image detection system. Fig. 8 illustrates how the platform analyzes uploaded photos to identify inappropriate or unsafe content before a complaint is submitted. This automated screening ensures that only valid and relevant images reach municipal authorities, protecting them from exposure to harmful material. The

feature also helps maintain the integrity of the platform by preventing misuse and keeping the reporting environment safe, respectful, and professional. By filtering content at an early stage, the system enhances both user trust and administrative efficiency.

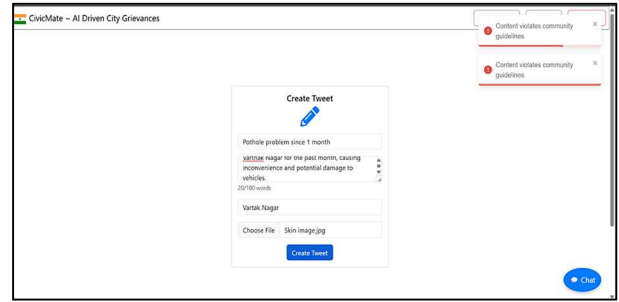


Fig. 8 NSFW Checker

D. AI-Powered Chatbot Interaction

To enhance user experience and offer instant support, CivicMate incorporates an AI-powered chatbot. Fig. 9 shows how the chatbot interacts with users by guiding them through the complaint submission process, answering frequently asked questions, and providing real-time updates on complaint status when needed. This conversational interface makes the platform more intuitive, especially for first-time users. It reduces dependency on manual navigation and ensures that help is available at any moment. By improving accessibility and engagement, the chatbot plays a key role in creating a smooth and user-friendly reporting journey.

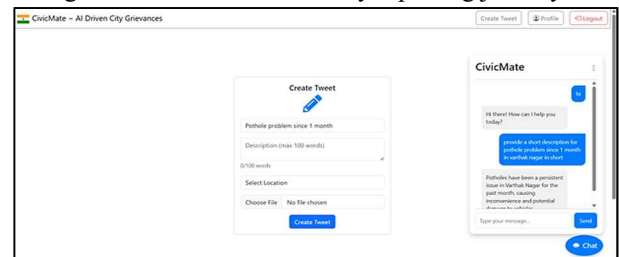


Fig. 9 Chatbot Interaction

E. Personalized User Profile and Community View

As Fig. 10 illustrates the user's personalized profile, where citizens can conveniently review all their submitted complaints and track the progress of each issue over time. The interface provides a clear history of actions and status updates, helping users stay informed and engaged. Additionally, the community view allows citizens to explore complaints reported by others in their locality, fostering greater awareness of common civic issues. Features such as upvoting enable users to support important concerns raised by neighbors, helping authorities identify high-impact problems.

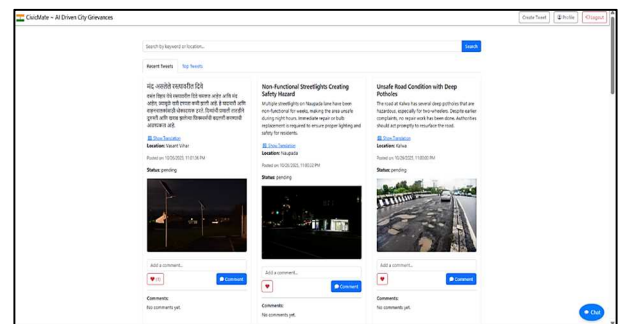


Fig. 10 Community Complaints View

F. Model Validation and Performance Evaluation

To evaluate the reliability of CivicMate's automated content-moderation pipeline, the NSFW detection and classification components were tested across multiple deep learning architectures. The results demonstrate high accuracy, fast inference, and practical deployment readiness for real-world civic reporting platforms.

1) *NSFW Detection Accuracy (Detector V2/V3)*: The NudeNet-based detection model, built on ResNet-101 and YOLO backbones and trained on a dataset of approximately 160,000 auto-labeled images, delivered strong performance across key metrics:

- Precision: 0.98
- Recall: 0.96
- F1-Score: 0.97

These results indicate that the system effectively filters inappropriate content while minimizing false positives and false negatives.

2) *Granular Classification and System Efficiency*: For fine-grained classification across 6–18 anatomical categories (e.g., exposed genitalia, breast, buttocks, torso), the system achieved:

- InceptionV3 Classification F1-Score: 0.950
- MobileNetV2 Classification F1-Score: 0.945

Latency evaluations demonstrated suitability for both cloud and edge deployment:

- High-Performance GPU Inference: ~0.024 sec (24.2 ms) per prediction.
- Edge/CPU Device Inference: ~0.41 sec per prediction.

These efficiency results confirm that the model can run smoothly on standard computing hardware while maintaining high accuracy.

A comparative analysis was carried out to evaluate CivicMate against traditional grievance redressal systems and other widely used digital platforms. Table I. summarizes the key features across these solutions, focusing on automation, transparency, user experience, and administrative capabilities. The results clearly show that CivicMate offers a more comprehensive, end-to-end approach by combining AI-driven content moderation, and community engagement features.

TABLE I. FEATURE COMPARISON OF GRIEVANCE REDRESSAL PLATFORMS

Feature / Parameter	Traditional Digital System	Snap Send Solve [5]	CitySolution Platform	CivicMate
Submission Method	Manual Text Image	Image + Location	Image Only	Text + Image + Auto-location
AI Auto-Categorization	Manual	Manual	Yes	Yes
AI Smart Assignment to Staff	No	No	No	Yes
Real-Time Status Tracking	No	Limited	No	Yes
Citizen Feedback	No	No	No	Yes

As shown in Table I, existing grievance redressal platforms typically excel in either technical accuracy or basic usability, but rarely in both. Most solutions lack fully automated complaint handling, robust content moderation, and meaningful mechanisms for citizen participation. CivicMate addresses these limitations by integrating AI-powered verification, automated categorization, real-time tracking, and community engagement features into a single unified platform. By combining intelligent automation with transparency and accessibility, CivicMate sets a new benchmark for modern urban grievance management systems.

VII. CONCLUSION

CivicMate is a smart and user-friendly platform that modernizes the civic grievance redressal process by empowering citizens and simplifying city-level management. It streamlines complaint submission through AI-based categorization, provides transparent real-time tracking, and encourages community participation. For authorities, the system offers an efficient dashboard with actionable insights, helping improve accountability and public trust. While the platform currently depends on image quality for accurate classification, future enhancements will focus on more advanced AI models for deeper contextual understanding and blockchain for secure, tamper-proof records. Overall, CivicMate supports smarter and more responsive urban governance.

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